Denominators of two basic hypergeometric ratios

A proof of MathDB problem 346880

Principia Math In Progress

September 12, 2026

Task 1: solve the original problem. We prove Conjecture 8.4 of Kim–Stanton [1], which is the MathDB question, and its companion Conjecture 8.3. These are Conjectures 7.3 and 7.2, respectively, in the earlier manuscript [2]. Only one MathDB problem is counted.

Let a, b, c, q be independent indeterminates and write

$$(u;q)_j = \prod_{i=0}^{j-1}(1-uq^i), \qquad F(a,b,c;z) = {}_2\phi_1(a,b;c;q,z) = \sum_{j\ge 0}\frac{(a;q)_j(b;q)_j}{(c;q)_j(q;q)_j}z^j.$$

Define formal series

$$R(z) = \frac{F(aq,b,cq;z)}{F(a,b,c;z)} = \sum_{n\ge 0} r_n z^n, \qquad S(z) = \frac{F(aq,bq,cq;z)}{F(a,b,c;z)} = \sum_{n\ge 0} s_n z^n.$$

Their constant coefficients are $r_0 = s_0 = 1$. Put

$$D_n(c,q) = \prod_{k=0}^{n}(1-cq^k)^{\lfloor (n+1)/(k+1)\rfloor}.$$

Theorem 1. *For every integer $n \ge 0$,*

$$\frac{D_n(c,q)}{1-c}\, r_n \in \mathbb{Z}[a,b,c,q], \qquad \frac{D_n(c,q)}{1-c}\, s_n \in \mathbb{Z}[a,b,c,q]. \tag{1}$$

The first assertion is the second coefficient-denominator conjecture recorded as MathDB 346880.

We first remove possible denominators involving q alone. Set

$$B = \mathbb{Z}[a,b,q], \quad A = B[c], \quad \mathcal{L} = A[(1-cq^j)^{-1} : j \ge 0], \quad K = \mathbb{Q}(a,b,q).$$

Lemma 2. *Every coefficient of R and S belongs to $\mathcal{L}$. Moreover,*

$$\mathcal{L} \cap K[c] = A. \tag{2}$$

Proof. Heine's contiguous relation, also given in [1, p. 13], is

$$F(aq,b,cq;z) - F(a,b,c;z) = \beta z F(aq,bq,cq^2;z), \qquad \beta = \frac{(a-c)(1-b)}{(1-c)(1-cq)}. \tag{3}$$

It can be checked directly in each coefficient: the relevant numerator simplification is

$$(1-au)(1-c) - (1-a)(1-cu) = (a-c)(1-u), \qquad u = q^j,$$

and $1-q^j$ cancels from $(q;q)_j$. Let σ substitute $(a,b,c) \mapsto (b,aq,cq)$, leaving q, z fixed. Symmetry of F in its first two parameters changes (3) into

$$R = \frac{1}{1-\beta z\,\sigma(R)}, \qquad r_n = \beta \sum_{j=0}^{n-1} r_j\, \sigma(r_{n-1-j}) \quad (n \ge 1). \tag{4}$$

Both $\beta \in \mathcal{L}$ and $\sigma(\mathcal{L}) \subseteq \mathcal{L}$ hold. Induction on n proves $r_n \in \mathcal{L}$.

A second elementary contiguous identity is

$$F(aq, bq, cq; z) - F(aq, b, cq; z) = \delta z F(aq^2, bq, cq^2; z), \qquad \delta = \frac{b(1-aq)}{1-cq}. \tag{5}$$

For a direct verification, in the coefficient of z^j the shift $b \mapsto bq$ contributes $b(1-q^j)/(1-b)$; cancellation in the shifted factorials gives the right side of (5). If τ substitutes $(a, b, c) \mapsto (aq, b, cq)$, then

$$S = R\big(1 + \delta z\, \tau(S)\big), \qquad s_n = r_n + \delta \sum_{j=0}^{n-1} r_j\, \tau(s_{n-1-j}). \tag{6}$$

Since τ also preserves $\mathcal{L}$, induction proves $s_n \in \mathcal{L}$.

Finally, write an element of $\mathcal{L}$ as U/V with $U \in A$ and V a finite product of powers of $1 - cq^j$. Its constant coefficient in c is $V(0) = 1$, so its Taylor series at $c = 0$ has all coefficients in B. If U/V is also a polynomial in $K[c]$, that Taylor series is finite. The polynomial therefore lies in $B[c] = A$, proving (2). □

Proof of Theorem 1. Let H denote either $F(aq, b, cq; z)$ or $F(aq, bq, cq; z)$, and put $F = F(a, b, c; z)$. We bound the poles, as functions of c, of the coefficients of

$$\frac{H}{(1-c)F}. \tag{7}$$

Work over $K = \mathbb{Q}(a, b, q)$. The points $c = q^{-k}$, $k \geq 0$, are distinct, and each factor $1 - cq^k$ is linear in c.

Fix $k \geq 1$ and put $\epsilon = 1 - cq^k$. The coefficient of z^j in F is regular at $c = q^{-k}$ for $j \leq k$, and has at most a simple pole for $j \geq k+1$. For H the corresponding threshold is $j = k$: its coefficients are regular for $j < k$ and have at most a simple pole thereafter. Thus, over the local ring of rational functions regular at this point,

$$\epsilon F = \epsilon U + z^{k+1} V, \qquad \epsilon H = \epsilon W + z^k X, \tag{8}$$

where U, W, V, X are formal series with regular coefficients, and $U(0) = W(0) = 1$. In particular U is a unit in the formal series ring over the local ring. Expand

$$\frac{H}{F} = \left(W + \frac{z^k X}{\epsilon}\right) \frac{1}{U} \sum_{\ell \geq 0} \frac{(-1)^\ell z^{(k+1)\ell} V^\ell}{\epsilon^\ell U^\ell}. \tag{9}$$

For each coefficient of z^n only finitely many summands contribute. The part involving W has pole order at most $\lfloor n/(k+1) \rfloor$. The other part vanishes if $n < k$; otherwise its pole order is at most

$$1 + \left\lfloor \frac{n-k}{k+1} \right\rfloor = \left\lfloor \frac{n+1}{k+1} \right\rfloor. \tag{10}$$

The factor $1 - c$ is a local unit because $k \geq 1$, so (10) is also the required bound for (7). In particular there is no pole when $k > n$.

The point $c = 1$ requires separate treatment. Now $\epsilon = 1 - c$, all coefficients of H are regular, and

$$\epsilon F = \epsilon + zV$$

with regular coefficients in V. Hence (7) becomes

$$\frac{H}{\epsilon + zV} = H \sum_{\ell \geq 0} \frac{(-1)^\ell z^\ell V^\ell}{\epsilon^{\ell+1}}. \tag{11}$$

Its coefficient of z^n has pole order at most $n + 1$, which is precisely the exponent prescribed at $k = 0$.

Lemma 2 shows that the coefficients of (7) lie in $\mathcal{L}$, so there are no other possible finite poles in c. Equations (10)–(11) show that D_n removes every such pole. The resulting rational function therefore lies in $K[c]$. It also lies in $\mathcal{L}$, so (2) places it in $A = \mathbb{Z}[a, b, c, q]$, as required. □

For example, the normalized polynomial for R at $n = 1$ is $(a-c)(1-b)$, and both normalized polynomials at $n = 0$ are 1. The proof is an identity over the field of rational functions; specializations are understood wherever the original series coefficients are defined, or through the resulting rational identities.

Task 2: determine the minimal denominators. Compute the exact generic pole orders of $r_n/(1-c)$ and $s_n/(1-c)$ at every $c = q^{-k}$. Determine when the exponents in D_n are attained and classify parameter specializations that lower them.

Task 3: treat arbitrary integral parameter shifts. For nonnegative integers u, v, w, study

$$F(aq^u, bq^v, cq^w; z)/F(a, b, c; z).$$

Determine which shifts admit normalizations in $\mathbb{Z}[a, b, c, q]$ and derive their optimal pole bounds.

Task 4: give explicit formulas for the normalized numerators. Construct finite sums or combinatorial models for $D_n r_n/(1-c)$ and $D_n s_n/(1-c)$ in which denominator cancellation has already been carried out. Use them to explain systematic factors and coefficient patterns.

Task 5: extend the method to higher basic hypergeometric series. Identify contiguous ratios of ${}_{r+1}\phi_r$ whose coefficients lie in an integral localization of their lower parameters, and establish denominator theorems by combining that property with local pole bounds.

References

[1] J. S. Kim and D. Stanton, *Three families of q-Lommel polynomials*, Séminaire Lotharingien de Combinatoire **87B** (2023), Article 8, 25 pp.; Conjectures 8.3–8.4. Published full text.

[2] J. S. Kim and D. Stanton, *Ratios of Hahn–Exton q-Bessel functions and q-Lommel polynomials*, arXiv:2006.08120v2 (2021), Conjectures 7.2–7.3. arXiv record.